DAY 06 – IPv4 ADDRESSING ASSIGNMENT

Name: Nethra M | Assignment Code: DAY06-IPV4 | Total Marks: 20 | Domain: Networking Fundamentals & Cybersecurity

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Part A: IPv4 Theory (10 Marks)

1. What is IPv4?

IPv4 stands for Internet Protocol Version 4. It is a network-layer protocol used to identify devices and route data packets between networks.

2. How many bits are in an IPv4 address?

An IPv4 address contains 32 bits.

3. What is an octet?

An octet is a group of 8 bits. Its decimal value ranges from 0 to 255.

4. How many octets are in an IPv4 address?

An IPv4 address contains four octets separated by dots. Example: 192.168.1.10.

5. What is a subnet mask?

A subnet mask identifies the network portion and host portion of an IPv4 address. Example: 255.255.255.0.

6. Difference between network portion and host portion?

Network portion: identifies the network or subnet. Host portion: identifies an individual device interface within that subnet.

7. What is a private IPv4 address?

A private IPv4 address is used within local networks, such as home, college, and office networks. It is not directly routable over the public internet.

8. Write all three private IPv4 ranges.

Private IPv4 range	CIDR notation
10.0.0.0 – 10.255.255.255	10.0.0.0/8
172.16.0.0 – 172.31.255.255	172.16.0.0/12
192.168.0.0 – 192.168.255.255	192.168.0.0/16

9. What is a public IPv4 address?

A public IPv4 address is an address that can be routed across the internet, subject to internet routing policies and address allocation rules.

10. What is the IPv4 loopback address?

127.0.0.1 is the commonly used IPv4 loopback address. It allows a computer to send network traffic back to itself for testing.

11. What is APIPA?

APIPA stands for Automatic Private IP Addressing. Windows may automatically assign an address from 169.254.0.0/16 when it cannot obtain a usable IPv4 address from a DHCP server.

12. What does 169.254.x.x commonly indicate?

It commonly indicates that a device has assigned itself an APIPA address because it did not receive a usable IPv4 address through DHCP.

13. What is a default gateway?

A default gateway is usually a router that forwards traffic from a local network to destinations outside that network.

14. Static vs dynamic IP addressing?

Static IP addressing: an IP address is configured manually or reserved to remain consistent. Dynamic IP addressing: an IP address is assigned automatically, usually by DHCP, and may change over time.

15. What is DHCP?

DHCP stands for Dynamic Host Configuration Protocol. It automatically assigns network settings, such as an IP address, subnet mask, default gateway, and DNS server addresses.

16. What is a network address?

A network address identifies a particular IPv4 subnet. In a conventional IPv4 subnet, all host bits are set to zero in the network address.

17. What is a broadcast address?

A broadcast address is used to send a packet to all hosts on a particular IPv4 subnet. In a conventional IPv4 subnet, all host bits are set to one in the broadcast address.

18. Why is IPv4 knowledge important in cybersecurity?

IPv4 knowledge helps cybersecurity professionals identify network devices, perform network mapping, understand subnetting, analyze network traffic, configure firewalls, troubleshoot connectivity, and investigate suspicious network activity.

Part B: Practical Identification (4 Marks)

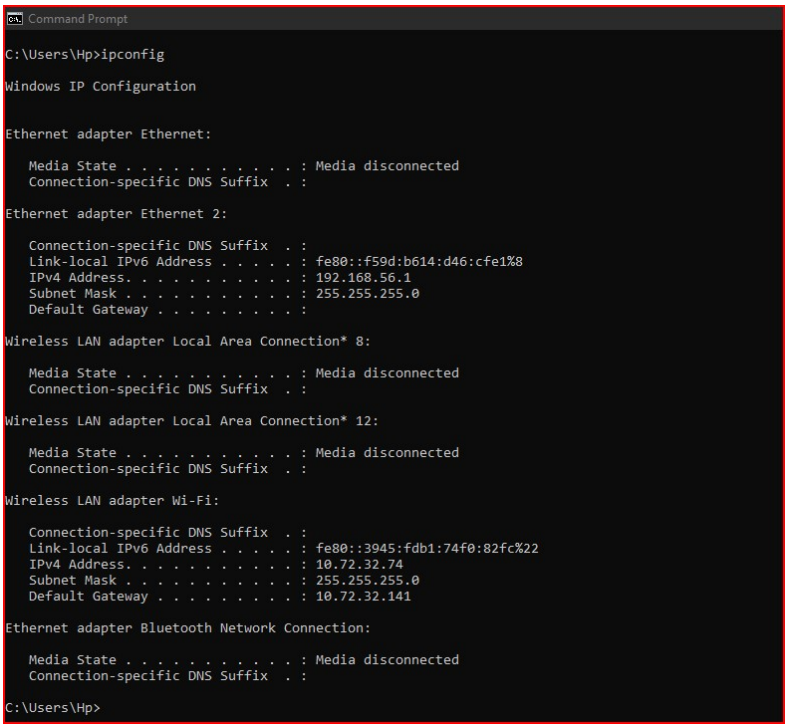
The plain *ipconfig* screenshot confirms the active connection is the **Wireless LAN adapter Wi-Fi** (the Ethernet adapter is disconnected, and Ethernet 2 is a virtual VirtualBox Host-Only adapter). The table below uses the connected Wi-Fi adapter's values, as the assignment requires.

Required detail	Your result
IPv4 address	10.72.32.74
Subnet mask	255.255.255.0
Default gateway	10.72.32.141
DNS server(s)	Not shown by plain *ipconfig* – run *ipconfig /all* and scroll to the Wi-Fi section to capture this
DHCP status	Not explicitly shown for Wi-Fi in the captured output – likely Enabled, typical for Wi-Fi, but not confirmed

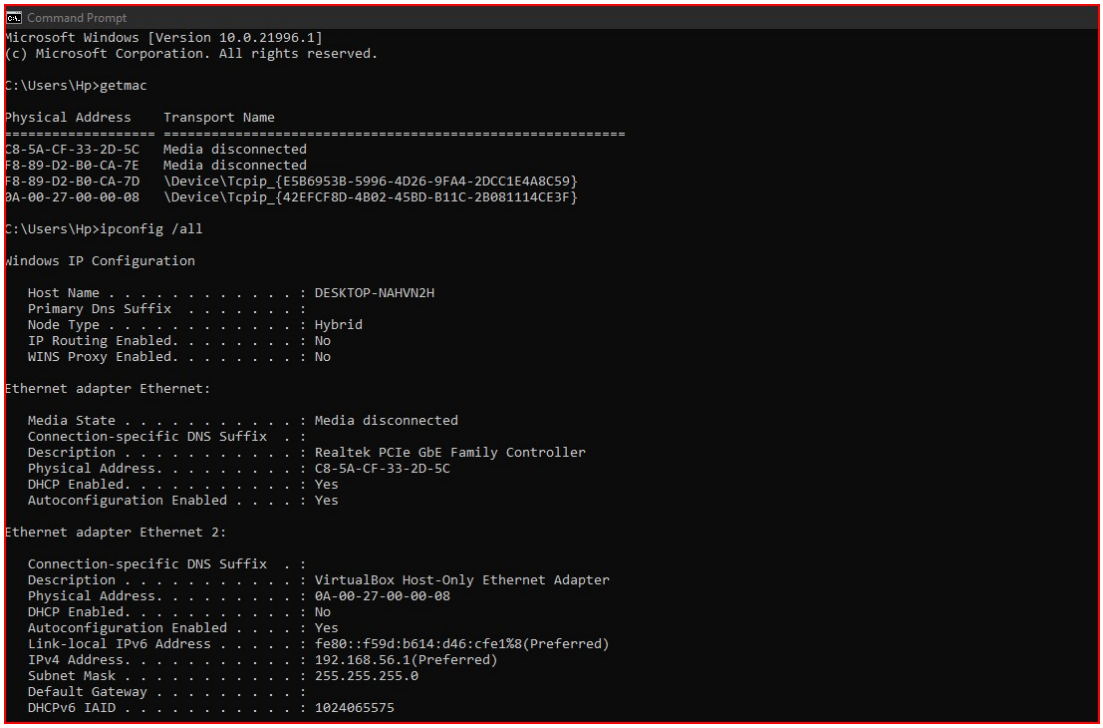
Note: *the Ethernet 2 (VirtualBox Host-Only) adapter seen earlier is a virtual adapter used by VirtualBox and is not the connection to the internet, so it is no longer used in this table. If you want the DNS server(s) and a confirmed DHCP status, run ipconfig /all and scroll down to the "Wireless LAN adapter Wi-Fi" section – send that screenshot and I'll complete the table.*

Part C: Screenshots (3 Marks)

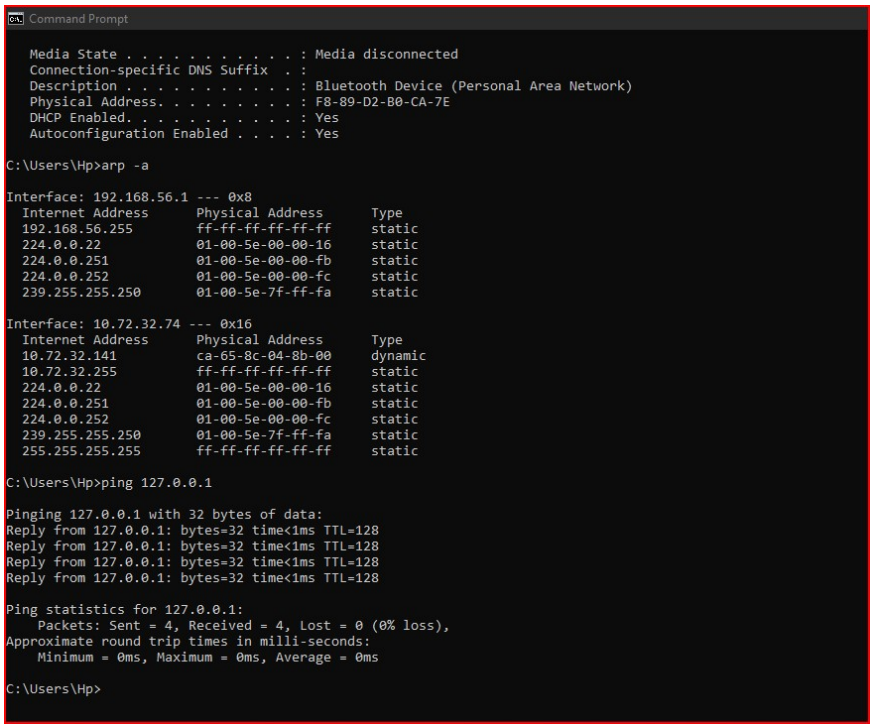
Screenshot 1 – ipconfig



Screenshot 2 – ipconfig /all



Screenshot 3 – ping 127.0.0.1



Confirms local TCP/IP loopback functionality: 4 packets sent, 4 received, 0% loss, round-trip time <1ms.

Part D: Report Quality (3 Marks)

- Name Nethra M and Day 06 appear on page 1.
- All 18 theory questions are answered.
- Practical details are filled in from the connected Wi-Fi adapter (DNS server(s) and confirmed DHCP status pending – see note above).
- All three screenshots are attached and readable.
- Formatting and spelling checked.

Conclusion

This assignment helped me understand IPv4 addressing, subnet masks, private and public IP ranges, APIPA, loopback addresses, default gateways, and DHCP. By practicing ipconfig, ipconfig /all, and ping 127.0.0.1, I learned how to inspect network configuration and test local TCP/IP functionality. These concepts provide a foundation for network troubleshooting and cybersecurity.